# Numerical-Text Surrogate Modelling and Multi-Objective Optimisation of Compact Patch Antennas

Module 2 of the Applied Vision Transformer Framework for Hybrid-Slotted DGS Antennas in 5G and Emerging 6G Networks - Proof-of-Concept Prototype

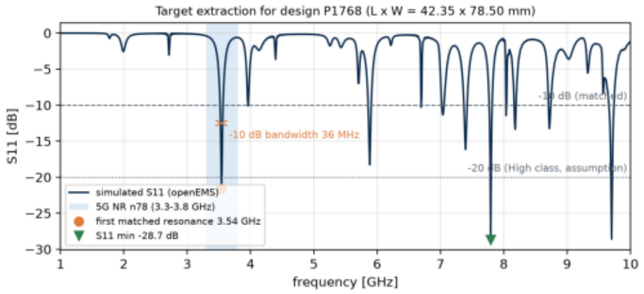
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## 1 Research objective

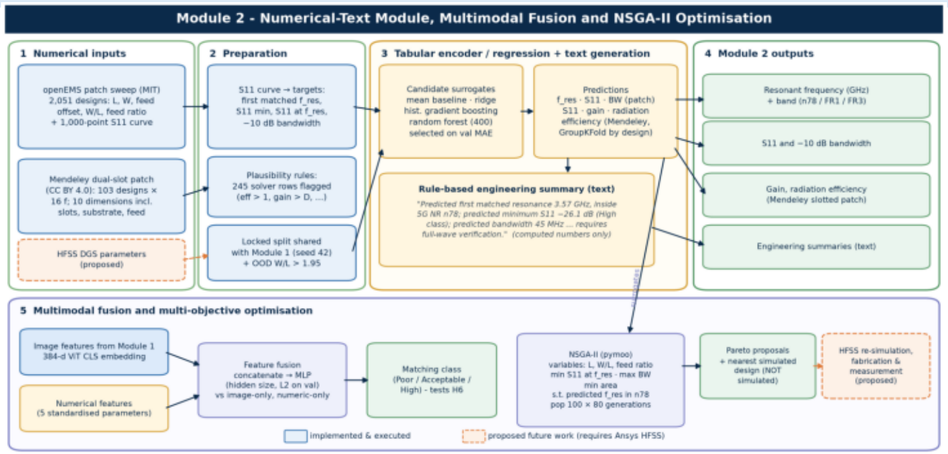
- How well do antenna parameters predict simulated resonance, S11, bandwidth, gain and efficiency?
- Do numbers add to Module 1's ViT image features? Can NSGA-II on surrogates propose n78 designs?
- Text summaries must never invent values.
- Proof of concept on public simulation data; Ansys HFSS was not available and was not run.

## 2 Data & preparation

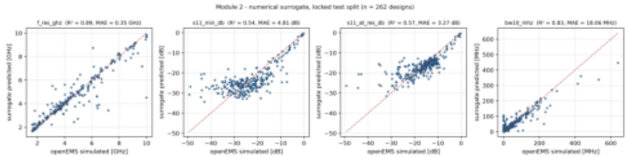
- openEMS patch sweep (Hugging Face, MIT): 2,051 designs; S11 curve -> 4 engineering targets
- Mendeley dual-slot patch v2 (CC BY 4.0): 103 x 16 f; 245 implausible solver rows flagged by a fixed rule
- Split shared with Module 1: 1,220 / 262 / 262 + OOD W/L > 1.95 (n = 307); GroupKFold for Mendeley
- Model selection on validation only; test scored once




## 3 Module 2 block diagram (implemented vs proposed)




## 4 Surrogate results (executed)




| Target (random forest) | Test R² | MAE | OOD R² |
|---|---|---|---|
| First matched resonance [GHz] | 0.89 | 0.35 | 0.44 |
| Minimum S11 [dB] | 0.54 | 4.81 | 0.13 |
| S11 at resonance [dB] | 0.57 | 3.27 | -1.54 |
| −10 dB bandwidth [MHz] | 0.83 | 18.06 | -3.72 |
| Mendeley S11 [dB] | 0.86 | 0.53 | - |
| Mendeley Gain [dBi] | 0.99 | 0.51 | - |
| Mendeley Radiation efficiency [ratio] | 0.97 | 0.03 | - |

*Mendeley: out-of-fold, 5-fold GroupKFold by design; gain and efficiency on plausible rows only*

**Takeaways**

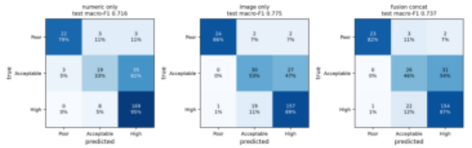
Resonance and bandwidth: well predicted

Matching depth: moderate (sharp impedance effect)

Wide OOD patches: fails -> search kept in range

Slot geometry halves the frequency-only error

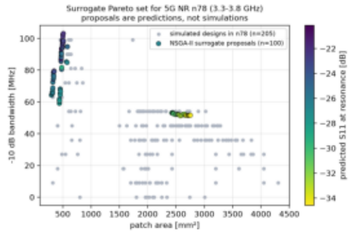
## 5 Multimodal fusion (H6)




Test macro-F1: fusion 0.737 | image-only 0.775 | numeric-only 0.716

*H6 not supported. ViT embeddings alone beat Module 1's own head (0.581)*

## 6 NSGA-II for 5G NR n78 (3.3-3.8 GHz)




100 Pareto proposals

area 299-2754 mm²

pred. BW 51-103 MHz

median 0.08 SD from a simulated design

**Predictions only - NOT simulated**

## 7 Outcomes & conclusion

| | | |
|---|---|---|
| **E1** | Surrogate beats mean baseline (all targets) | **Met** |
| **E2** | Slot geometry adds to frequency | **Met** |
| **H6** | Fusion beats image-only (pre-registered) | **Not supported** |
| **E3** | Feasible n78 proposals in range | **Met** |

- Surrogates are only trustworthy where they were trained
- Next: space-filling HFSS DGS sweep + re-simulation

**Fast, traceable surrogate screening of compact patch antennas for 5G / 6G - simulation evidence only, proposals not HFSS-validated**

Data: LaBash et al. (2025) arXiv:2505.18188 [MIT] | Mihad, Islam & Sojib (2024) Antenna Parameters Dataset v2, doi:10.17632/3gxr2vvd9n.2 [CC BY 4.0]
Random forest: Breiman (2001) | NSGA-II: Deb et al. (2002) | pymoo: Blank & Deb (2020) | Code: https://github.com/rathiabeth-sudo/902---Module-2